

Lab Exercise #10

Task 2: Write three Assembly language procedures to

- **Clear the screen. with the name CLRSCR,**
- **Set the cursor at top left corner of the screen, with the name SETCRSR,**
- **Display the table of any number from 2 to 9, with the name DISPTBL**

Call these procedures in a main program.

Code:

TITLE LAB 10 ACTIVITY 2 ; Title of the program

.MODEL SMALL ; Memory model for the program
 .STACK 100H ; Stack size

.DATA
 MSG DB 'Table: \$' ; Define a string message

.CODE

; Procedure to clear the screen

```
CLRSCR PROC
    MOV AH, 00H ; Function to set video mode
    MOV AL, 03H ; Video mode 3 (80x25 color text)
    INT 10H ; Video interrupt to set video mode

    XOR AX, AX ; Set AX to 0
    MOV BH, 07H ; Page number (0 to 7)
    MOV CX, 0 ; Upper-left corner row
    MOV DX, 184FH ; Lower-right corner row and column
    INT 10H ; Video interrupt to set cursor position
    RET ; Return from the procedure
CLRSCR ENDP
```

; Procedure to set the cursor at the top-left corner of the screen

```
SETCRSR PROC
    MOV AH, 02H ; Function to set cursor position
    MOV BH, 0 ; Video page number (0 for default)
    MOV DH, 0 ; Row (top-left corner)
    MOV DL, 0 ; Column (top-left corner)
    INT 10H ; Video interrupt to set cursor position
    RET ; Return from the procedure
SETCRSR ENDP
```

; Procedure to display the table of a specified number

```
DISPTBL PROC
    MOV BL, '0' ; Initialize BL to '0'

    ; Display message
    MOV AH, 09H ; Function to print a string
    LEA DX, MSG ; Load the offset address of the message
```

INT 21H ; DOS interrupt to print the message

MOV CX, 10 ; Set the limit for the table (e.g., 10 times)

DISPTBL_LOOP:

; MOVE TO THE NEXT LINE

MOV AH, 2 ; DOS function to move the cursor

MOV DL, 0AH ; Linefeed character

INT 21H ; DOS interrupt

; MOVE CURSOR TO THE BEGINNING OF THE LINE

MOV AH, 2 ; DOS function to move the cursor

MOV DL, 0DH ; Carriage return character

INT 21H ; DOS interrupt

; Display current number

MOV AH, 02H ; DOS function to display a character

MOV DL, '2' ; Display the number 2

INT 21H ; DOS interrupt

; Display multiplication symbol

MOV AH, 02H ; DOS function to display a character

MOV DL, '*' ; Display the multiplication symbol

INT 21H ; DOS interrupt

; Display current iteration

MOV AH, 02H ; DOS function to display a character

MOV DL, BL ; Display the constant value

INT 21H ; DOS interrupt

; Display equal sign

MOV AH, 02H ; DOS function to display a character

MOV DL, '=' ; Display the equal sign

INT 21H ; DOS interrupt

; Calculate and display the result

MOV AH, 00H ; Clear AH register

MOV AL, '2' ; Clear AL and set it to '2'

AND AL, 0FH ; Mask high bits to get the numeric value

AND BL, 0FH ; Mask high bits of BL

MUL BL ; Multiply AX by BL

AAM ; ASCII adjust AX to unpack binary-coded decimal

OR AX, 3030H ; Convert result to ASCII in AX

MOV BX, AX ; Move the ASCII value to BX

; Display high digit of the result

MOV AH, 02H ; DOS function to display a character

MOV DL, BH ; Move the high digit of the result to DL

INT 21H ; DOS interrupt

```
; Display low digit of the result
MOV AH, 02H      ; DOS function to display a character
MOV DL, BL       ; Move the low digit of the result to DL
INT 21H          ; DOS interrupt

INC BL           ; Increment BL for the next iteration

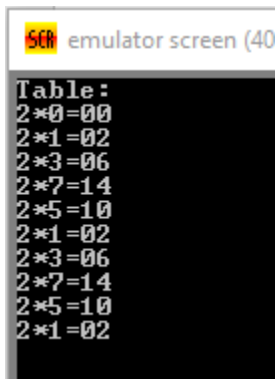
; Move to the next iteration
LOOP DISPTBL_LOOP ; Continue the loop until CX reaches the limit
RET              ; Return from the procedure
DISPTBL ENDP

MAIN PROC
MOV AX, @DATA    ; Load the data segment address to AX
MOV DS, AX       ; Initialize DS register with the data segment address

CALL CLRSCR      ; Call procedure to clear the screen
CALL SETCRSR     ; Call procedure to set the cursor at the top-left corner
CALL DISPTBL     ; Call procedure to display the table of a specified number

MOV AH, 4CH      ; DOS function to exit the program
INT 21H          ; DOS interrupt to exit

MAIN ENDP
END MAIN
```

Output:

```
SCH emulator screen (40)
Table :
2*0=00
2*1=02
2*3=06
2*7=14
2*5=10
2*1=02
2*3=06
2*7=14
2*5=10
2*1=02
```

